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COPULA ACTIVE SUBSPACES II: ERROR DECOMPOSITION, A POSTERIORI ESTIMATION, AND SHARPNESS OF THE BOUNDS*

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Abstract.
The Copula Active Subspaces (CAS) method of Part I estimates a non-Gaussian observation-noise density $\pi_{\boldsymbol{n}}$ from residual samples. Its KL divergence $\mathcal{D}_{\mathrm{KL}}(\pi_{\boldsymbol{n}} \,\|\, \widehat{\pi}_{\boldsymbol{n}})$ admits an exact decomposition into marginal, projection and reduced-density terms and bounds the posterior KL divergence averaged over the true predictive distribution. We derive a priori bounds for these terms, together with estimates or bounds from residual samples. Weighted nearest-neighbour entropy estimates are used for the marginal and reduced-density terms, with asymptotic standard errors in exact rank coordinates. For the projection error, the score is fitted on one subsample, and its mean squared magnitude outside the retained subspace is estimated on the other. Corrections for score and rank-transform errors then give a theoretical upper bound with probability at least $1-\delta$. Tests on three known noise laws in dimension $d = 20$ use the estimated rank transform. In 11 of 12 combinations of example and sample size, the sum of the two estimated terms and the trace bound exceeds the reference divergence after averaging over the estimated models. In these tests, score-estimation error is the largest contribution to looseness. The overestimate from the Gaussian log-Sobolev inequality alone is at most a factor of 1.98.

Key words. copula, active subspace, score matching, dimension reduction, Hermite polynomials, Bayesian inference problems, non-Gaussian observation error

MSC codes. 62G07, 62G20, 62H05, 62B10, 60E15, 62F15

1. Introduction. Approximating the observation-noise density changes the posterior in a Bayesian inference problem. The noise-density KL divergence bounds the resulting posterior KL divergence, averaged over the true predictive distribution. Part I [10] estimates a non-Gaussian noise density $\pi_{\boldsymbol{n}}$ from residual samples, and tests the estimate against known noise laws. Such a comparison is unavailable in applications. We therefore consider the error

$$\mathcal{D}_{\mathrm{KL}}(\pi_{\boldsymbol{n}} \,\|\, \widehat{\pi}_{\boldsymbol{n}}), \tag{1.1}$$

of the CAS estimate $\widehat{\pi}_{\boldsymbol{n}}$, estimating its individual contributions from the residual samples.

The decomposition in Proposition 2.1 assigns one error term to each estimation step of CAS: (1.1) equals $\mathcal{E}_{\mathrm{marg}} + \mathcal{E}_{\mathrm{proj}} + \mathcal{E}_{\mathrm{dens}}$. The marginal term $\mathcal{E}_{\mathrm{marg}}$ consists of d one-dimensional divergences and two corrections; $\mathcal{E}_{\mathrm{dens}}$ is a divergence between r-dimensional laws. For the projection error $\mathcal{E}_{\mathrm{proj}}$, the rank-$r$ reduction admits a bound in terms of the copula-score variance outside the retained subspace. This is the Gaussian-reference bound of certified dimension reduction [36].

A direct entropy estimate of (1.1) would involve the differential entropy of $\pi_{\boldsymbol{n}}$ in dimension d. For densities supported on $[0,1]^d$ in Lipschitz balls of smoothness $s = 2$, the minimax root-mean-square error has order $(N \log N)^{-2/(d+2)} + N^{-1/2}$ [19,

*This is Part II of a two-part work. The companion paper *Copula Active Subspaces I: A Score-Covariance Method for Reduced-Order Non-Gaussian Density Estimation* defines the estimator analyzed here.

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Thm. 1]. The decomposition reduces the required entropy calculations to dimensions 1 and r. No entropy estimate is needed for the bound on $\mathcal{E}_{\mathrm{proj}}$.

The a priori bounds in Section 3 depend on the rank r, the multi-index sets and the sample size N. Normalizing the fitted joint density introduces a correction in the marginal bound. The bound minus this correction tends to zero as $N \to \infty$, under the stated assumptions on the marginal estimates and rank-transform error. The bounds on $\mathcal{E}_{\mathrm{proj}}$ and $\mathcal{E}_{\mathrm{dens}}$ keep approximation, regularization and constraint terms that do not vanish as $N \to \infty$. In the examples of Section 6, $\mathcal{E}_{\mathrm{proj}}$ is the largest term for $N \geq 12{,}500$, and the Gaussian log-Sobolev bound overestimates it by a factor between 1.64 and 1.98. We also apply gradient descent to the tighter bound of [24]. The subspace moves by $\|\sin\Theta\|_F \leq 0.0004$ from the top-r eigenbasis of $\boldsymbol{C}$ used by CAS. Most of the remaining looseness of the a priori bound on these examples is due to the Hermite truncation of the score (Subsection 6.4).

For the a posteriori analysis in Section 4, we estimate $\mathcal{E}_{\mathrm{marg}}$ and $\mathcal{E}_{\mathrm{dens}}$ by weighted nearest-neighbour entropy estimators. Under the stated assumptions these attain the parametric rate in exact rank-Gaussianized coordinates. To bound $\mathcal{E}_{\mathrm{proj}}$, we fit the score and rank transform on half of the sample. The other half estimates the mean squared magnitude of the fitted score outside the retained subspace. Under the stated assumptions, a theoretical upper bound for this quantity holds with probability at least $1-\delta$. It becomes a bound on $\mathcal{E}_{\mathrm{proj}}$ after adding corrections for errors in the score estimate and rank transform. The increment between two nested multi-index sets estimates the Hermite truncation error under a saturation assumption. It does not estimate the sampling or regularization errors.

Section 2 defines the estimator and decomposes its error. The three numerical examples are reported in Section 6.

2. The estimator and its exact error decomposition. The two stages of CAS estimate the density $\pi_{\boldsymbol{n}}$ of the noise $\boldsymbol{n} \in \mathbb{R}^d$ from N residual samples (Part I [10], Algorithm 3.1). For marginal densities π_i with CDFs F_i, the componentwise map $\mathcal{Z}(\boldsymbol{x})_i := \Phi^{-1}(F_i(x_i))$ produces standard Gaussian marginals (2.1), with Φ the standard normal CDF. The transformed noise $\boldsymbol{Z} := \mathcal{Z}(\boldsymbol{n})$ has law $\pi_{\boldsymbol{Z}}$. Replacing F_i by the rank-based $\widehat{F}_i$ of Part I defines $\widehat{\mathcal{Z}}$. The marginal densities are estimated separately by Gaussian-kernel estimates $\widehat{\pi}_i$ [31].

Let γ_k denote the standard Gaussian density on $\mathbb{R}^k$. The copula density is $c^Z := \pi_{\boldsymbol{Z}}/\gamma_d$. Its score $\mathcal{S} := \nabla \log c^Z$ has second-moment matrix $\boldsymbol{C} := \mathbb{E}_{\pi_{\boldsymbol{Z}}}[\mathcal{S}\mathcal{S}^\top]$ (2.2), referred to as the score covariance. With eigenvalues ordered as $\lambda_1(\boldsymbol{C}) \geq \cdots \geq \lambda_d(\boldsymbol{C})$, a top-$r$ eigenbasis $\boldsymbol{V}_r^{\boldsymbol{C}}$ spans the copula active subspace.

Stage 1 fits the copula score by Hermite score matching [22] in the span of the gradients of multivariate Hermite polynomials H_α. The polynomials are orthonormal in $L^2(\gamma_d)$ and indexed by $\Lambda := \Lambda_{K_1,q_1}$. The set $\Lambda_{K,q}$ contains indices with $1 \leq |\alpha|_1 \leq K$ and at most q nonzero entries, except for the linear indices e_i (3.2). This fit defines $\widehat{\mathcal{S}} := \nabla \sum_{\alpha\in\Lambda} \widehat{\theta}^d_\alpha H_\alpha$ (3.3), with coefficient vector $\widehat{\boldsymbol{\theta}}^d := (\widehat{\theta}^d_\alpha)_{\alpha\in\Lambda}$. From the empirical second-moment matrix $\widehat{\boldsymbol{C}}$ of $\widehat{\mathcal{S}}$ (3.5), we choose a top-r eigenbasis $\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}}$. The superscript distinguishes this basis from a principal-component basis of $\boldsymbol{Z}$.

A basis $\boldsymbol{V}_r \in \mathrm{St}(r,\mathbb{R}^d)$ and its orthonormal completion $\boldsymbol{V}_\perp$ separate the retained coordinates $\boldsymbol{U} := \boldsymbol{V}_r^\top \boldsymbol{Z}$ from the discarded coordinates $\boldsymbol{W} := \boldsymbol{V}_\perp^\top \boldsymbol{Z}$. The conditional law of $\boldsymbol{W}$ given $\boldsymbol{U}$ is denoted by $\pi_{\boldsymbol{W}|\boldsymbol{U}}$. Define the reduced factor $c_r^U := \pi_{\boldsymbol{U}}/\gamma_r$ and its logarithm $\mathcal{L}_r := \log c_r^U$. Part I (Subsection 2.1) uses this factor to define the reduced model $\pi_{\boldsymbol{n}}(\boldsymbol{x};\boldsymbol{V}_r) := c_r^U(\boldsymbol{V}_r^\top \mathcal{Z}(\boldsymbol{x}))\prod_i \pi_i(x_i)$ (2.4). For the fitted estimator, we use $\boldsymbol{V}_r = \widehat{\boldsymbol{V}}_r^{\boldsymbol{C}}$ in these definitions.

Stage 2 uses the multi-index set $\Lambda_r := \Lambda_{K_2,q_2}$ in r variables to fit $\mathcal{L}_r(\boldsymbol{u};\widehat{\boldsymbol{\theta}}^r) := \sum_{\alpha\in\Lambda_r}\widehat{\theta}^r_\alpha H_\alpha(\boldsymbol{u})$. The coefficient vector $\widehat{\boldsymbol{\theta}}^r := (\widehat{\theta}^r_\alpha)_{\alpha\in\Lambda_r}$ solves the regularized and constrained score-matching problem (3.9). The Gram matrix $\widehat{\boldsymbol{A}}_r$, right-hand side $\widehat{\boldsymbol{b}}_r$ and centering matrix $\widehat{\boldsymbol{B}}$ have population counterparts $\boldsymbol{A}^\star$, $\boldsymbol{b}^\star$ and $\boldsymbol{B}^\star$, respectively, in exact coordinates. Dividing the exponential by $\widehat{Z}_r := \mathbb{E}_{\gamma_r}[e^{\mathcal{L}_r(\cdot;\widehat{\boldsymbol{\theta}}^r)}]$ produces the reduced factor $\widehat{c}^U_r := e^{\mathcal{L}_r(\cdot;\widehat{\boldsymbol{\theta}}^r)}/\widehat{Z}_r$ and the normalized density $\widehat{\pi}_{\boldsymbol{U}} := \widehat{c}^U_r\gamma_r$. Composing this factor with the fitted rank transform and multiplying by the marginal estimates requires another normalization, by $\widehat{Z}_{\mathrm{comp}}$. Thus the joint estimate (3.1) takes the form

$$\widehat{\pi}_{\boldsymbol{n}}(\boldsymbol{x}) := \widehat{Z}^{-1}_{\mathrm{comp}}\,\widehat{c}^U_r(\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}\top}\widehat{\mathcal{Z}}(\boldsymbol{x}))\prod_i \widehat{\pi}_i(x_i),$$

$$\widehat{Z}_{\mathrm{comp}} := \int_{\mathbb{R}^d}\widehat{c}^U_r(\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}\top}\widehat{\mathcal{Z}}(\boldsymbol{x}))\prod_i \widehat{\pi}_i(x_i)\,d\boldsymbol{x},$$

with copula factor $\widehat{c}^Z(\boldsymbol{z}) := \widehat{c}^U_r(\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}\top}\boldsymbol{z})$. The normalizing constant $\widehat{Z}_{\mathrm{comp}}$ equals 1 when each $\widehat{F}_i$ is the distribution function of $\widehat{\pi}_i$. In Cas, $\widehat{\mathcal{Z}}$ uses the empirical distribution functions and the $\widehat{\pi}_i$ are kernel density estimates, so this condition fails. In the notation of Part I, $\widehat{Z}_{\mathrm{comp}} = \widehat{Z}_{\boldsymbol{n}}/\widehat{Z}_r$, since $\widehat{c}^U_r$ already includes the factor $\widehat{Z}_r^{-1}$. The normalizing constants are evaluated by quasi-Monte Carlo (Appendix G). Hats denote estimates and unhatted symbols their exact counterparts. The symbols $\mathrm{Gr}(r,\mathbb{R}^d)$ and $\mathrm{St}(r,\mathbb{R}^d)$ denote the Grassmann and Stiefel manifolds [1], and $\Theta(\boldsymbol{V},\boldsymbol{V}')$ is the vector of principal angles between the spans of two bases [8].

Marginal assumption. Throughout we assume (M): $\mathbb{E}_{\pi_{\boldsymbol{n}}}[|\log(\pi_i/\widehat{\pi}_i)|] < \infty$ for every i, and $\mathbb{E}[\sum_i \mathcal{D}_{\mathrm{KL}}(\pi_i \,\|\, \widehat{\pi}_i)] = O(\sqrt{d}\,(\log N)^{K_2/2}N^{-1/2})$. The expectation is taken over repeated estimation samples. The assumed rate requires the marginal density error to decay at least as fast as the rank-transform contribution under (T) (Subsection 3.1). We do not derive this rate for the Gaussian-kernel estimates.

Proposition 2.1 (Decomposition of the KL divergence by estimation stage). *Let $\widehat{\pi}_{\boldsymbol{n}}$ be the estimator above, with $\boldsymbol{U} := \widehat{\boldsymbol{V}}_r^{\boldsymbol{C}\top}\boldsymbol{Z}$. Under Assumption 2.1 of Part I and the integrability condition of* (M)*, every term below is finite and, for each realization of the estimation sample,*

(2.1)

$$\mathcal{D}_{\mathrm{KL}}(\pi_{\boldsymbol{n}} \,\|\, \widehat{\pi}_{\boldsymbol{n}}) = \underbrace{\textstyle\sum_{i=1}^d \mathcal{D}_{\mathrm{KL}}(\pi_i \,\|\, \widehat{\pi}_i) + R_N + \log \widehat{Z}_{\mathrm{comp}}}_{\mathcal{E}_{\mathrm{marg}}\ \textit{(the marginals)}} + \underbrace{\mathcal{D}_{\mathrm{KL}}\big(\pi_{\boldsymbol{n}} \,\|\, \pi_{\boldsymbol{n}}(\,\cdot\,;\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}})\big)}_{\mathcal{E}_{\mathrm{proj}}\ \textit{(the subspace)}}$$

$$+ \underbrace{\mathcal{D}_{\mathrm{KL}}\big(\pi_{\boldsymbol{U}} \,\|\, \widehat{\pi}_{\boldsymbol{U}}\big)}_{\mathcal{E}_{\mathrm{dens}}\ \textit{(the reduced density)}},$$

where $R_N := \mathbb{E}_{\pi_{\boldsymbol{n}}}[\log \widehat{c}^Z(\mathcal{Z})] - \mathbb{E}_{\pi_{\boldsymbol{n}}}[\log \widehat{c}^Z(\widehat{\mathcal{Z}})]$ is the difference between the expected log copula factors evaluated at the exact and estimated rank transforms. The normalizing constant $\widehat{Z}_{\mathrm{comp}}$ is defined above, and $\mathcal{E}_{\mathrm{proj}} = \mathbb{E}_{\pi_{\boldsymbol{U}}}[\mathcal{D}_{\mathrm{KL}}(\pi_{\boldsymbol{W}|\boldsymbol{U}} \,\|\, \gamma_{d-r})]$.

Appendix A proves this decomposition at the estimated basis $\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}}$, without the eigenbasis of $\boldsymbol{C}$. Neither R_N nor $\log \widehat{Z}_{\mathrm{comp}}$ has a prescribed sign, so $\mathcal{E}_{\mathrm{marg}}$ can be negative.

3. A priori error bounds.

3.1. The marginals. For a fixed dictionary of degree K_2 we assume the moment bounds (T),

(3.1)

$$\mathbb{E}\|\widehat{\boldsymbol{\theta}}^r\|^2 = O(1), \qquad \mathbb{E}\,\mathbb{E}_{\pi_{\boldsymbol{n}}}\Big[(1+\|\mathcal{Z}\|+\|\widehat{\mathcal{Z}}\|)^{2(K_2-1)}\|\widehat{\mathcal{Z}}-\mathcal{Z}\|^2\Big] = O\bigg(\frac{d(\log N)^{K_2}}{N}\bigg).$$

where the outer expectation is over the estimation sample and the inner one over a fresh residual. A mean-value bound for the polynomial $\widehat{\mathcal{L}}_r$ and the Cauchy–Schwarz inequality give $\mathbb{E}|R_N| = O(\sqrt{d}\,(\log N)^{K_2/2}N^{-1/2})$ (Appendix A). Combining this with (M) gives

$$\mathbb{E}[\mathcal{E}_{\mathrm{marg}} - \log \widehat{Z}_{\mathrm{comp}}] \;=\; O\big(\sqrt{d}\,(\log N)^{K_2/2}N^{-1/2}\big), \tag{3.2}$$

and $\mathcal{E}_{\mathrm{marg}} - \log \widehat{Z}_{\mathrm{comp}} = O_p(\sqrt{d}\,(\log N)^{K_2/2}N^{-1/2})$. The constants depend on d and the dictionary. We compute the remaining correction $\log \widehat{Z}_{\mathrm{comp}}$ for each estimate and do not bound it a priori.

3.2. The subspace. The Stage-1 bounds follow from the next inequality, in which $\mathcal{S}'$ is an arbitrary score field.

Proposition 3.1 (Stage-1 bounds). *Let $\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}}$ be the estimated Stage-1 basis of* (3.5) *and $\widehat{\boldsymbol{P}}_r := \widehat{\boldsymbol{V}}_r^{\boldsymbol{C}}\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}\top}$. Under Assumption 2.1 of Part I, for every score field $\mathcal{S}' \in L^2(\pi_{\boldsymbol{Z}};\mathbb{R}^d)$ with second-moment matrix $\boldsymbol{C}' := \mathbb{E}_{\pi_{\boldsymbol{Z}}}[\mathcal{S}'\mathcal{S}'^{\top}]$,*

$$\sqrt{2\,\mathcal{E}_{\mathrm{proj}}} \;\le\; \|\mathcal{S}-\mathcal{S}'\|_{L^2(\pi_{\boldsymbol{Z}})} \;+\; \sqrt{\mathrm{Tr}\big((\boldsymbol{I}-\widehat{\boldsymbol{P}}_r)\boldsymbol{C}'\big)}. \tag{3.3}$$

For $\mathcal{S}' = \mathcal{S}$, the score-error term in (3.3) *vanishes. Squaring the remaining inequality and dividing by two yields the oracle bound*

$$\mathcal{E}_{\mathrm{proj}} \;=\; \mathcal{D}_{\mathrm{KL}}\big(\pi_{\boldsymbol{n}}\,\|\,\pi_{\boldsymbol{n}}(\,\cdot\,;\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}})\big) \;\le\; C_1 \;:=\; \tfrac{1}{2}\,\mathrm{Tr}\big((\boldsymbol{I}-\widehat{\boldsymbol{P}}_r)\boldsymbol{C}\big), \tag{3.4}$$

The oracle bound uses half the trace of $\boldsymbol{C}$ over the discarded subspace. For comparison, let $\boldsymbol{P}_r^{\boldsymbol{C}} := \boldsymbol{V}_r^{\boldsymbol{C}}\boldsymbol{V}_r^{\boldsymbol{C}\top}$ project onto the top-r eigenspace of $\boldsymbol{C}$. Subtracting and adding the trace at this projector separates the rank and suboptimality contributions:

$$C_1 \;=\; \underbrace{\tfrac{1}{2}\textstyle\sum_{i>r}\lambda_i(\boldsymbol{C})}_{\textit{rank}} \;+\; \underbrace{\tfrac{1}{2}\big[\mathrm{Tr}(\boldsymbol{P}_r^{\boldsymbol{C}}\boldsymbol{C}) - \mathrm{Tr}(\widehat{\boldsymbol{P}}_r\boldsymbol{C})\big]}_{\Delta_{\mathrm{subopt}}\,\ge\,0}, \tag{3.5}$$

where the rank term is fixed by $\boldsymbol{C}$ and r. The remaining term Δ_{subopt} is of order $\|\sin\Theta(\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}}, \boldsymbol{V}_r^{\boldsymbol{C}})\|_F^2$. It is zero whenever $\mathrm{span}(\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}}) = \mathrm{span}(\boldsymbol{V}_r^{\boldsymbol{C}})$, and only in this case if $\lambda_r(\boldsymbol{C}) > \lambda_{r+1}(\boldsymbol{C})$.

To prove Proposition 3.1, apply the triangle inequality in $L^2(\pi_{\boldsymbol{Z}};\mathbb{R}^d)$ to the Gaussian-reference bound (2.5) of Part I (Appendix A). This bound is [36, Cor. 2.10] specialized to a standard Gaussian reference. In the score-ratio-matching stability result of [5, Thm. 3.3], the divergence at the top-r eigenspace of $\boldsymbol{C}'$ is bounded by $\|\mathcal{S}-\mathcal{S}'\|^2_{L^2(\pi_{\boldsymbol{Z}})} + \sum_{i>r}\lambda_i(\boldsymbol{C}')$. This follows from (3.3) at that eigenprojector after applying $(a+b)^2 \le 2a^2+2b^2$. Our bound retains the square root and applies to the estimated projector $\widehat{\boldsymbol{P}}_r$.

Projecting $\mathcal{S}$ onto the gradient space $\mathcal{F}_\Lambda$ in $L^2(\pi_{\boldsymbol{Z}})$ defines $\mathcal{S}_\Lambda$, its second-moment matrix $\boldsymbol{C}_\Lambda$, and the top-r projector $\boldsymbol{P}_r^{\Lambda\star}$. Because $\widehat{\boldsymbol{C}}$ is computed from the penalized